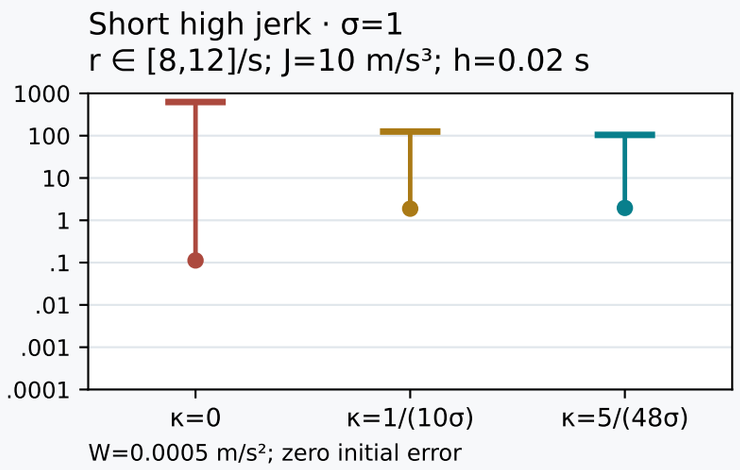
A tracking bound is not a measured tracking error

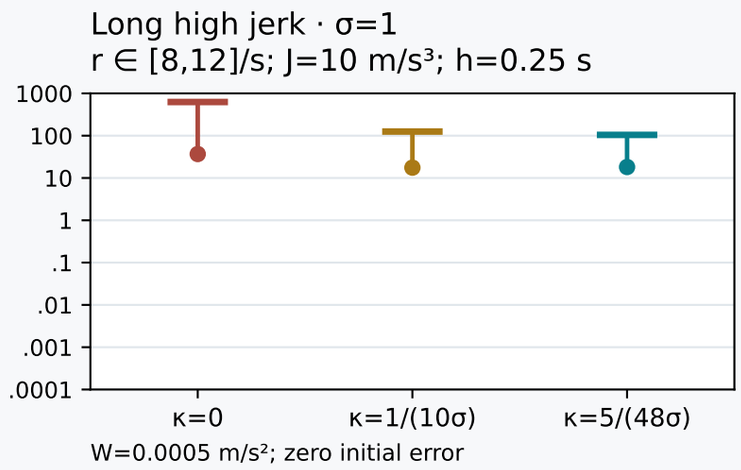
Actual scalar plant; continuous-state feedback; no saturation. Numerical traces are evidence, not certification.

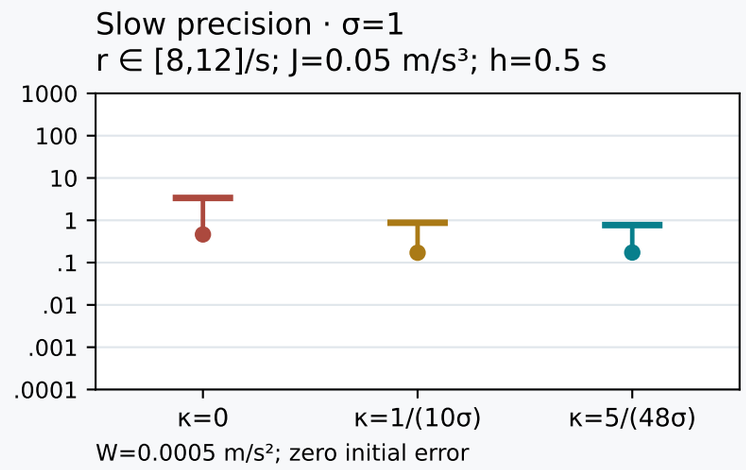
Same reference and disturbance across coefficients. σ = 10 assumes a 10× faster physical response band.

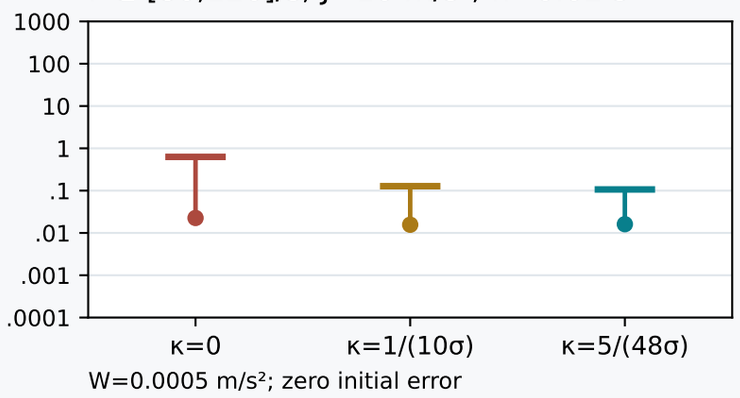
Acceleration FF only · Nominal-rate jerk FF · Reciprocal-midpoint FF

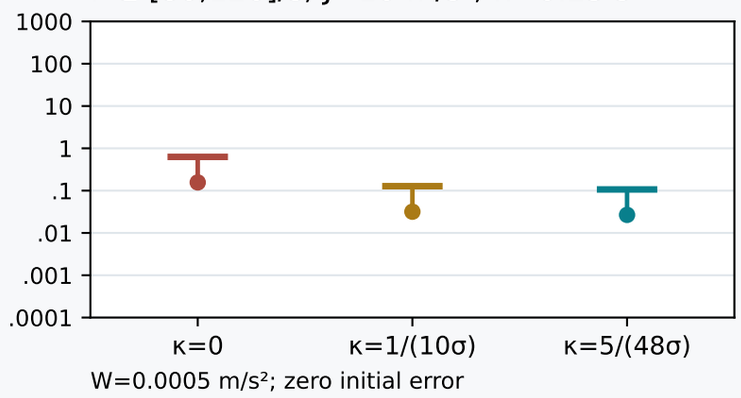
Top panels: cap = sufficient radius; dot = largest sampled error across four declared rate signals. Log scale in mm.

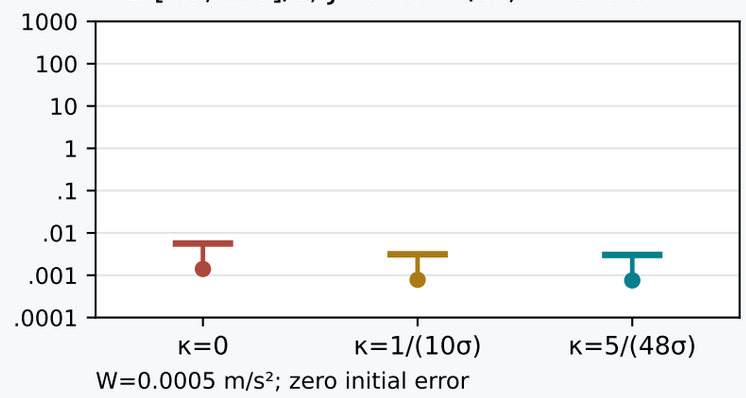




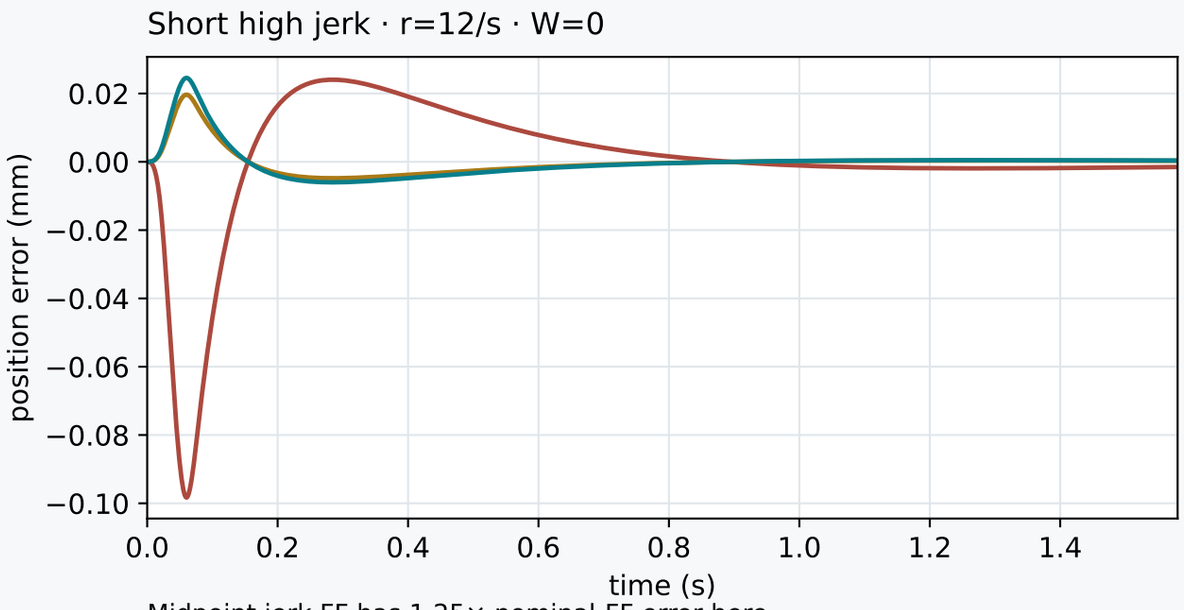




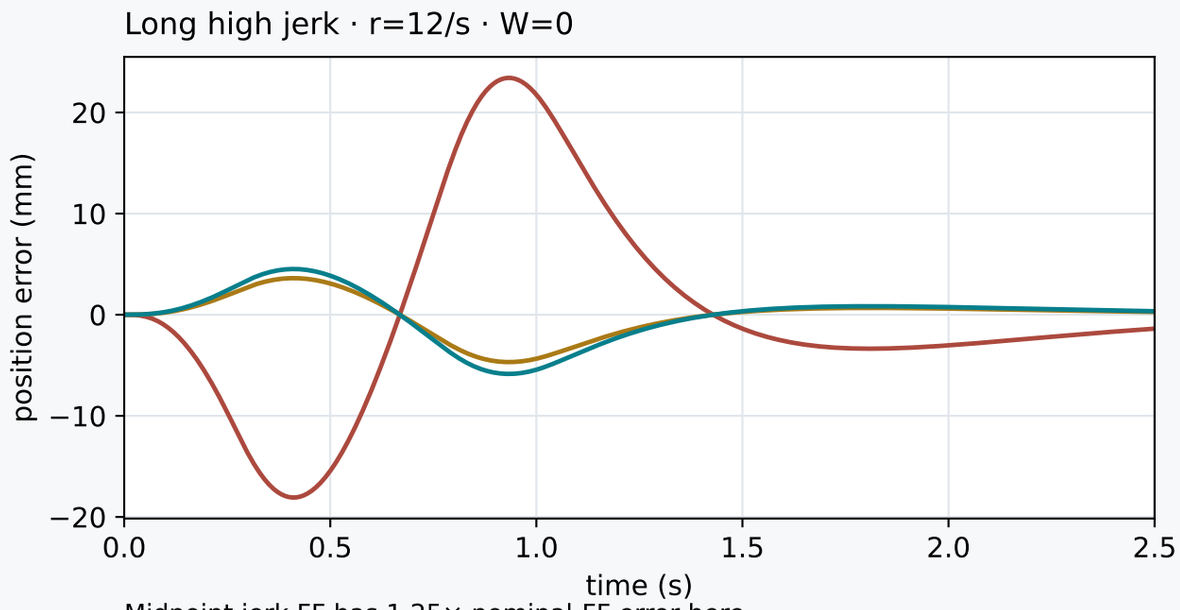




A smaller certificate radius does not make a coefficient best for every trace



Midpoint jerk FF has 1.25× nominal-FF error here.
Linear model, identical forcing; no universal ranking.



Midpoint jerk FF has 1.25× nominal-FF error here.
Linear model, identical forcing; no universal ranking.

The sufficient tube covers the stated model uncertainty; plotted peaks sample only the declared signals.

The short-pulse gap exposes conservatism. Faster response is a plant assumption, not a gain-only upgrade.

No robot, VLA, contact, sampled-feedback, or hardware performance claim. JSON/CSV contain all 144 cases and refinement checks.